# 1 Periodic Functions and Their Graphs

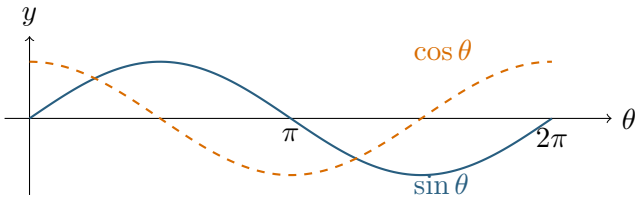
**Central question.** How does rotation become a repeating signal, and how can we recover the motion from a graph?

## 1.1 The circle makes a wave

For a point traveling around the unit circle with angle $\theta$, its vertical coordinate is $y = \sin\theta$ and its horizontal coordinate is $x = \cos\theta$. Each revolution restores the same coordinates. Hence

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

The *period* is the smallest positive input change that repeats the full function. Both sine and cosine have period $2\pi$. Sine starts at 0 and initially rises; cosine starts at 1.




## 1.2 Amplitude and vertical translation

Consider $y = A\sin x + D$. Its midline is $y = D$, its amplitude is $|A|$, and its range is $[D - |A|, D + |A|]$. A negative $A$ reflects the wave across the midline.

**Example 1.1.** The function $y = 3\sin x - 2$ has amplitude 3, midline $y = -2$, maximum 1, and minimum $-5$.

## 1.3 Period and horizontal scaling

For $y = \sin(Bx)$ with $B \neq 0$, one full cycle occurs when $Bx$ changes by $2\pi$; the period is $T = 2\pi/|B|$.

**Example 1.2.** The graph of $y = \cos(4x)$ repeats every $\pi/2$ units of input.

When $x$ denotes time measured in seconds, the quantity $|B|$ is angular frequency in radians per second, and ordinary frequency is $f = 1/T = |B|/(2\pi)$ cycles per second.

## 1.4 Phase shift

A useful general form is

$$\boxed{y = A\sin(B(x - C)) + D}.$$

Here $C$ is the horizontal shift, $D$ the midline, $|A|$ the amplitude, and $2\pi/|B|$ the period. We should distinguish $B(x - C)$ from $Bx - C$: factoring first prevents an incorrect phase shift.

**Example 1.3.** For $y = 2\sin(3x - \pi) + 1 = 2\sin(3(x - \pi/3)) + 1$, the period is $2\pi/3$, the shift is $\pi/3$ to the right, the amplitude is 2, and the midline is 1.

### 1.5 Five landmark points

Rather than sketching a sinusoid by guesswork, divide one period into four equal intervals. For $A > 0$, sine begins on the midline, rises to a maximum after one quarter-period, returns to the midline, reaches a minimum, then returns to the midline. A negative amplitude reverses the rises and falls.

**Try it.** Find five landmark points of $y = -2\cos(\pi(x-1)) + 3$, showing one entire period.

### 1.6 Models of repeating motion

A Ferris wheel has radius 9 m, its center is 11 m above ground, and it makes one revolution every 30 s. If a passenger begins at the rightmost position and travels counterclockwise, their height is

$$h(t) = 11 + 9\sin\left(\frac{2\pi}{30}t\right).$$

Its amplitude is 9 m, midline is 11 m, and period is 30 s. A sinusoidal model can describe a wheel, a spring, a tide, or a musical tone, provided its parameters actually correspond to the chosen physical quantities.

### 1.7 Tangent and reciprocal graphs

The tangent function repeats after $\pi$, not $2\pi$, because both sine and cosine change sign after a half turn and their ratio remains the same:

$$\tan(x+\pi) = \tan x.$$

Tangent has vertical asymptotes at $x = \pi/2 + k\pi$. The secant and cosecant graphs arise as reciprocals of cosine and sine; their breaks occur wherever the respective denominator is zero.

### 1.8 Exercises

1. Find the amplitude, midline, period, and range of $y = -4\sin(2x) + 1$.

2. Rewrite $y = 3\cos(5x - \pi) + 2$ in shifted form and identify the phase shift.

3. Give five landmark points for $y = 2\sin(x - \pi/2)$.

4. Explain why $y = \sin(2x)$ completes two periods on $[0, 2\pi]$.

5. A wheel of radius 6 m rotates once every 24 seconds. Its center is 8 m above the ground. Model the rider's height when starting at the rightmost point.

6. Identify the vertical asymptotes of $y = \tan(2x)$ on $[0, 2\pi)$.

7. Sketch one period of $y = 2\sec x$ and indicate all discontinuities.

8. Explain how you can determine amplitude and midline directly from observed maximum and minimum values.